Artifact Requirements

This file lists the hardware, software, storage, and time requirements for the artifact accompanying *Suborbital and Orbital Real-Time Localization of Gamma-Ray Bursts via Utility-Guided Coordination*. See `INSTALL.md` for installation and data-download instructions and `README.md` for the experiment workflows.

Evaluation paths

The artifact supports three evaluation paths:

1. **Visualize the distributed results.** Reproduce Figures 4--11 from the precomputed CSV files and notebook included with the artifact.
2. **Run the complete experiment workflow.** Regenerate the training and evaluation results used by the utility-guided pipeline.
3. **Optionally rerun the Figure 4 mapping benchmark.** Measure the five Intel x86-64 implementations and/or the three Jetson ARM64 implementations.

Figures 1--3 are explanatory diagrams and are not generated by the artifact.

Hardware and operating system

- Operating system: Linux.
- Complete training and evaluation workflow: ARM64 (`aarch64`).
- Reference ARM64 platform: NVIDIA Jetson Orin NX with eight ARM Cortex-A78AE v8.2 CPU cores, CPU frequency fixed at 1.5 GHz, and 16 GB DRAM. The GPU is not used.
- Optional Intel measurements in Figure 4: Intel/AMD x86-64.
- Reference Intel platform for Figure 4: Intel Xeon Gold 5118 at 2.3 GHz with 256 GB DRAM.
- Storage: at least 65 GB after extraction, plus temporary space for the 16.2 GB downloaded archive.
- Non-commodity peripherals: none.

The storage may be located anywhere with sufficient capacity. The instructions use a configurable `ARTIFACT_ROOT` variable and do not require the repository, dependencies, or datasets to be placed under `${HOME}`.

Timing results are platform-sensitive. Functional execution on another compatible machine does not imply matching the measurements reported for the reference platforms.

Software

The following tools and libraries are required for the complete workflow:

- Bash and standard Linux command-line tools;
- Git;
- `wget` or `curl`, `tar`, and `md5sum`;
- Conda or Miniconda;
- Python 3.12 and the packages specified by `cosipy-312-deps.yaml`;
- Jupyter Notebook;

- **CMake 3.26 or later** (required to build the current HDF5 source);
- Make and a C++17 compiler with OpenMP support;
- LEMON Graph Library 1.3.1;
- the HDF5 C library; the installation also builds the optional C++ and high-level libraries;
- HighFive headers;
- the Bitshuffle HDF5 plugin supplied by the Python `hdf5plugin` package; and
- the `tsmap-artifact` branch of COSIpy.

COSIpy is not included as a directory in the initial GRB-Search-Pipeline checkout; `INSTALL.md` clones and installs it separately.

For the optional Intel/x86-64 measurements in Figure 4, use `cosipy-312-intel.yml`. This second environment is needed only for that cross-platform benchmark and is not required for the complete ARM64 training and evaluation workflow.

The two Conda YAML files are the authoritative Python package specifications for their respective architectures.

Data and network access

- Source repository: approximately 200 MB.
- Compressed `transients.tar.gz` archive: approximately 16.2 GB.
- Extracted detector models and transient datasets: approximately 25--30 GB.
- Maps produced by complete experiment runs: approximately 30 GB.

Network access is required during installation to obtain repositories, dependencies, and the Zenodo data archive. Once these resources are installed, the experiments can run offline.

Expected skills

Reviewers should be comfortable using a Linux shell, Conda environments, environment variables, CMake-based C++ builds, and Jupyter notebooks. No specialized astronomy knowledge is required to execute the artifact.

Approximate execution time

- Visualizing the distributed results: a few minutes.
- Full training phase: approximately 18 hours.
- Full evaluation phase: approximately 21 hours.
- Optional Intel Figure 4 benchmark: approximately 20 minutes.
- Optional Jetson Figure 4 benchmark: approximately 10 minutes.

Actual times depend on the platform, storage system, and available CPU resources.